Convolutional Neural Networks (CNN)

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rom Pixels to Predictions, An Intuitive Approach

The Problem

The Biological Inspiration

End-to-End CNN Workflow

Technical Demo

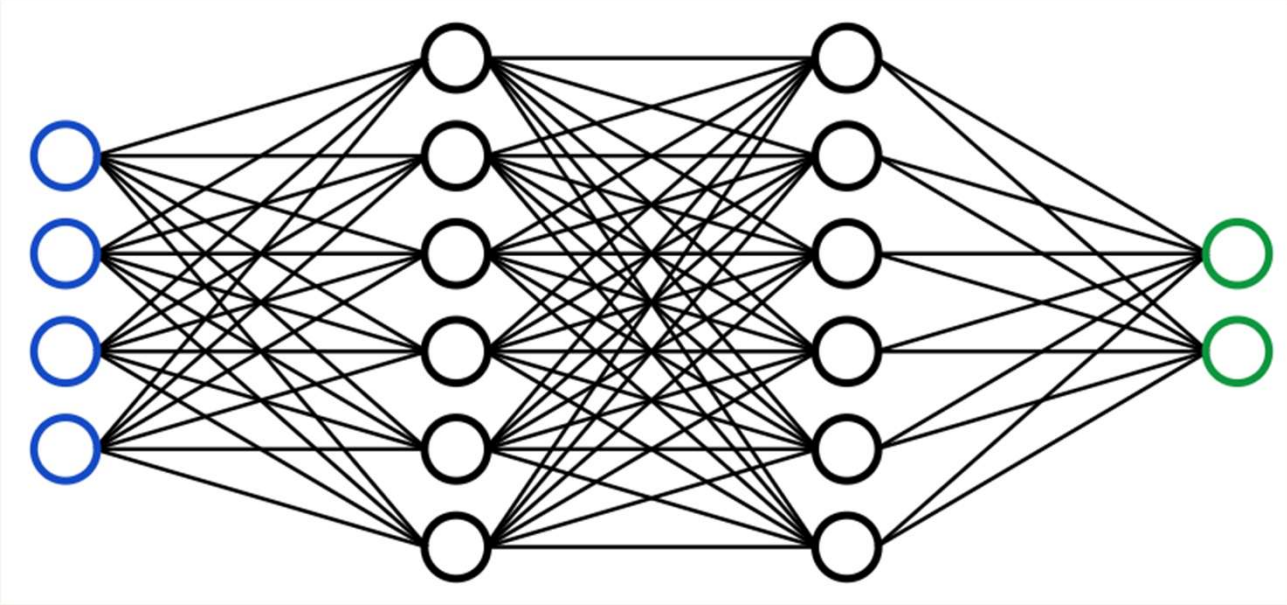
Future Implementations

lentifying the Problem In CNNs

Computers process images as matrices of numbers

Spatial relationships between pixels

Massive blowup in the number of network parameters



NNs Emulate Biological Findings

Visual patterns

Local receptive fields and shared weights

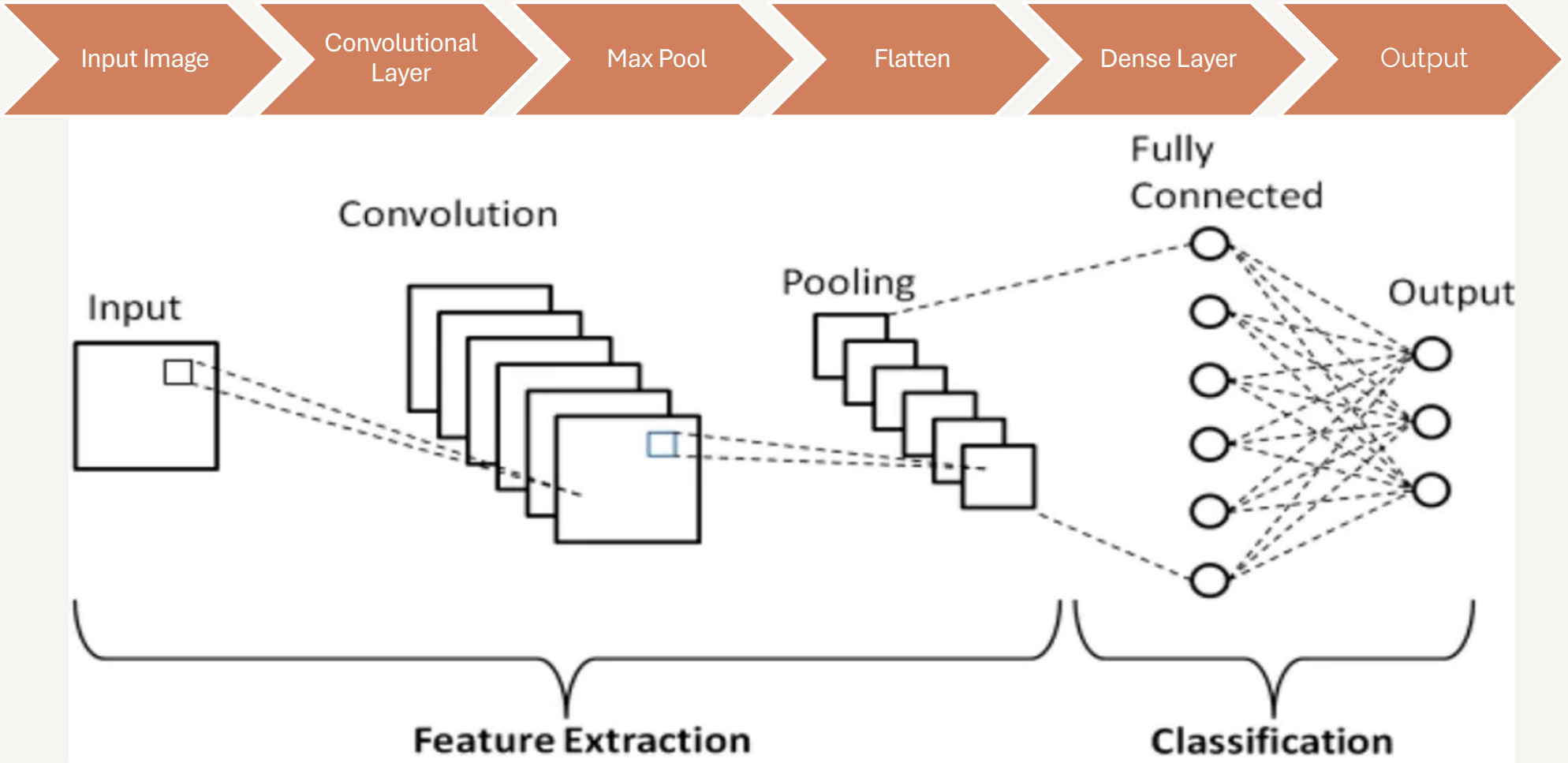
The Visual Cortex (Hubel and Weisel)

Cognitron & Neocognitron (Fukushima)

Optimized Pattern Recognition (LeCun et al.)

Modern Day Computer Vision

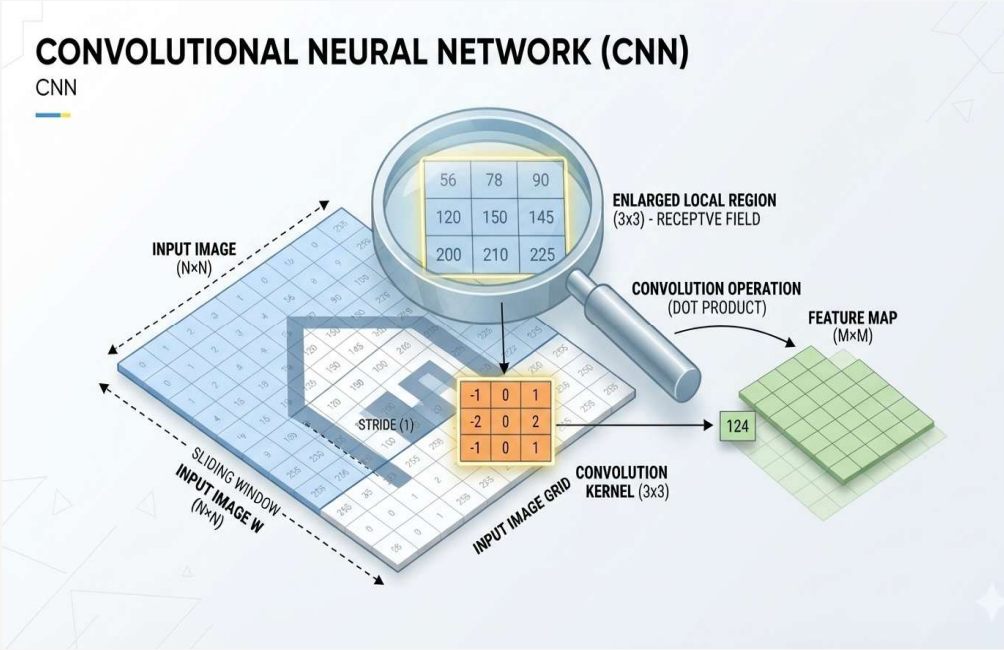
he End-to-End CNN Workflow



Source: Phung, Van & Rhee, Eun. (2019). A High-Accuracy Model Average Ensemble of Convolutional Neural Networks for Classification of Cloud Image Patches on Small Datasets. Applied Sciences. 9.4500.10.3390/app9214500.

onvolution: Divide and Conquer

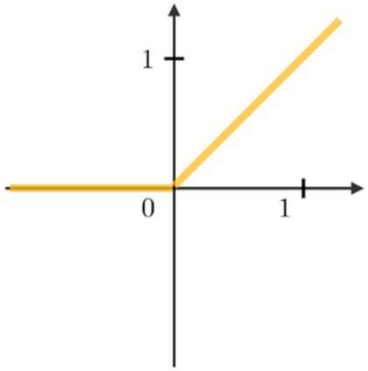
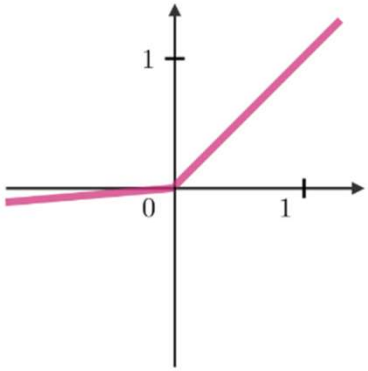
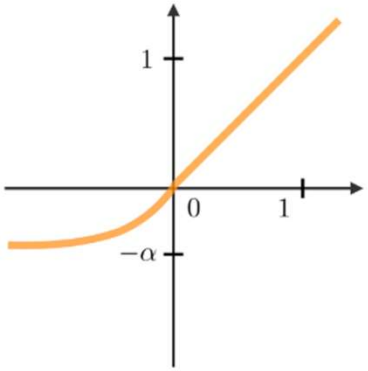
- Local Receptive Fields
- The Filtering Process, *"The Magnifier Glass"*
- Feature Maps and visual features
- Convolutions, mapping local connectivity and learning distinct spatial patterns



Note. Image generated using the prompt "put a magnifying glass over the input," by Gemini, 2026 (google.com).

on-Linearity for Real-World Data

Feature Map transformations to add complexity

ReLU	Leaky ReLU	ELU
$g(z) = \max(0, z)$	$g(z) = \max(\epsilon z, z)$ with $\epsilon \ll 1$	$g(z) = \max(\alpha(e^z - 1), z)$ with $\alpha \ll 1$
		
• Non-linearity complexities biologically interpretable	• Addresses dying ReLU issue for negative values	• Differentiable everywhere

Source: Amidi, A., & Amidi, S. (n.d.). CS 230 - Convolutional Neural Networks cheatsheet. Stanford University. https://stanford.edu/~shervine/teaching/cs-230/cheatsheet-convolutional-neural-networks/#activation

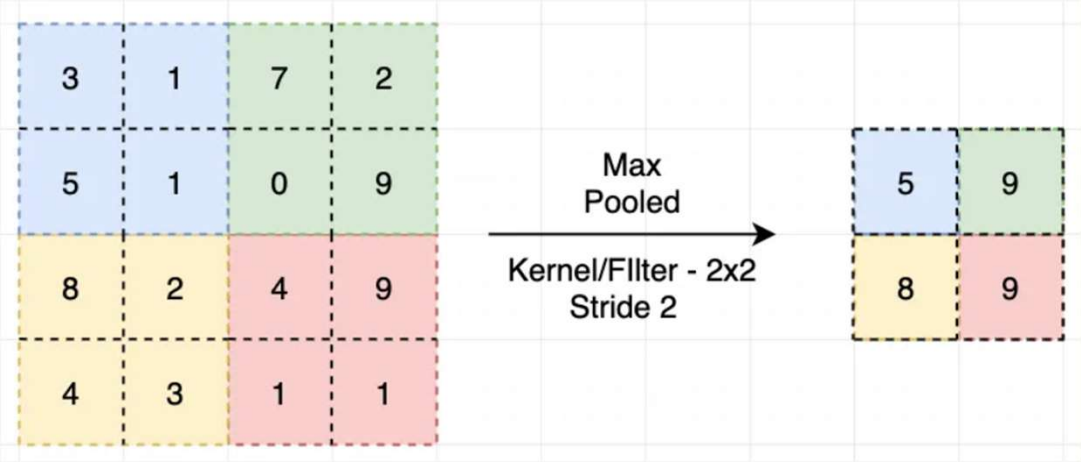
Different Perspectives: Pooling / Downsampling

Reduces spatial resolution, typically via Max Pooling

Deeper layers used to scan broader, more abstract configurations

From Simple lines and edges to complex shapes and objects

"The Magnifying Glass"



Source: Rana, K. (2020, April 20). *Pooling layer — Short and simple*. In Plain English. https://plainenglish.io/artificial-intelligence/pooling-layer-beginner-to-intermediate-fa0dbdce80eb

he Learning Engine & The Loss Function

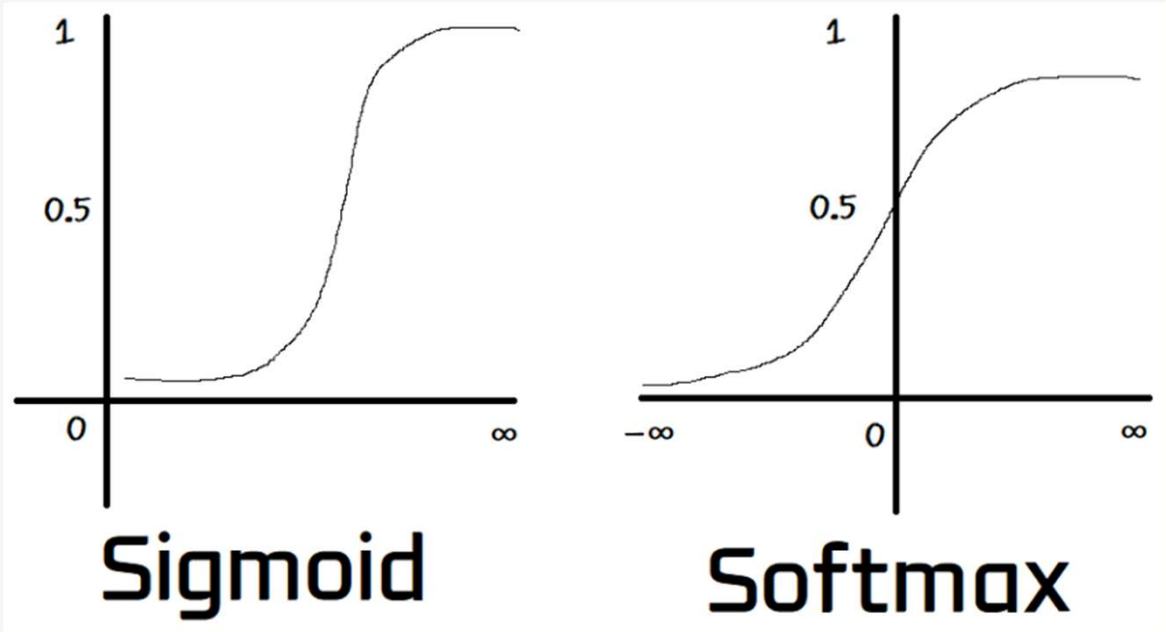
The Dense Layers

Optimization

- Weight changes
- Loss function

Probability Distribution

Final Output



Source: Chiro-J. (2022, August 9). #13 Softmax Function. Velog. velog.io

echnical Implementation Demo

Training dataset of 60,000 images

28x28 grayscale images

10 digits

Test set of 10,000 images

Pixel values range from 0-255

uture Implementations

Project Overview & Dataset

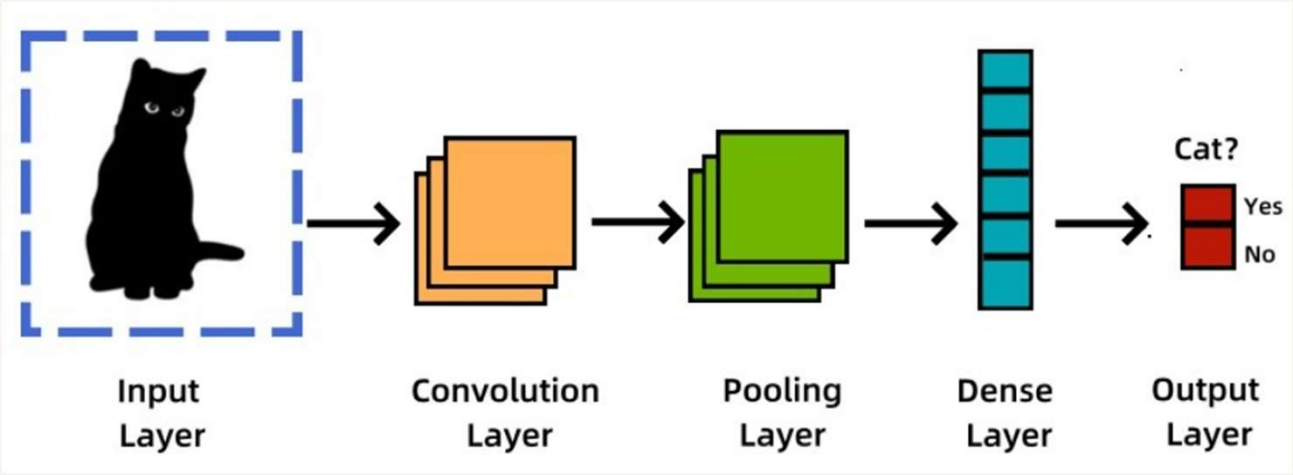
- **Multi-class classification** utilizing Convolutional Neural Network (CNN) modeling
- **Data Source: COVID-19 Radiography Database** (Kaggle), compiled by Qatar University, University of Dhaka, and international partners
- **Dataset Composition: 21,165 total Chest X-Ray (CXR) images** across four distinct classes:
    - **Normal / Healthy:** 10,192 images
    - **Lung Opacity (Non-COVID):** 6,012 images
    - **COVID-19 Positive:** 3,616 images
    - **Viral Pneumonia:** 1,345 images

Methodological Extensions

Can the model serve as a reliable second opinion?

How close are the COVID-19 and viral pneumonia predictions?

uestions?


Cat?
Yes
No
Input Layer
Convolution Layer
Pooling Layer
Dense Layer
Output Layer

eferences

Fukushima, K. (1980). NeoCognitron: a self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. In *Biological Cybernetics* (Vols. 36–36, pp. 193–202) [Journal-article]. Springer-Verlag. https://www.rctn.org/bruno/public/papers/Fukushima1980.pdf

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